AI Alone Won't Transform HealthCare

It Requires an Information Architecture

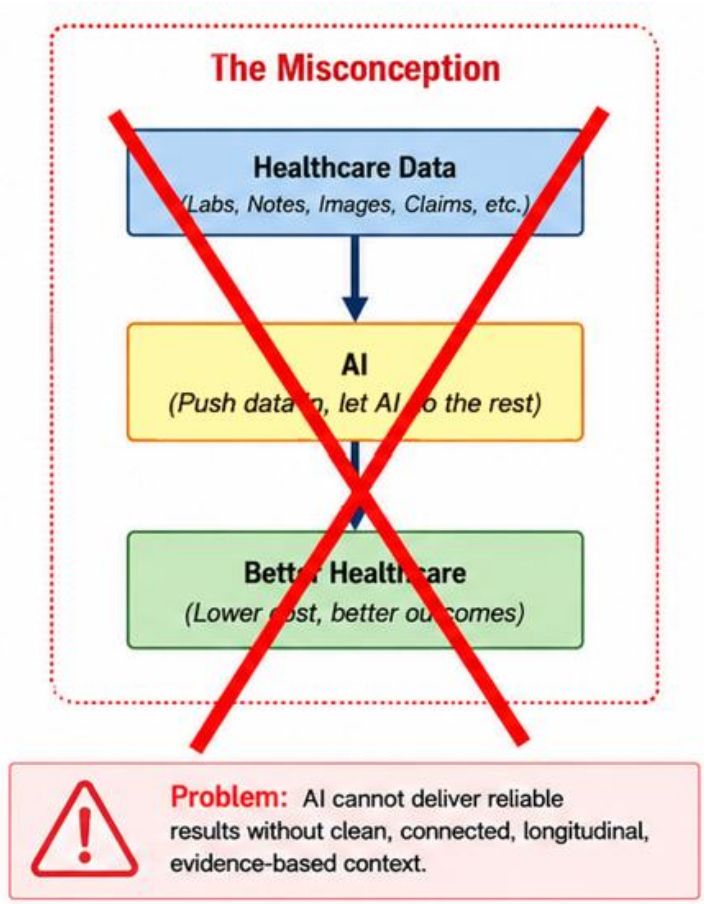
The more I studied my proposal to use AI to streamline healthcare, the harder It became.

My proposal for a national longitudinal patient history didn't begin with technology. It began with observations.

Like many families, mine has repeatedly encountered redundant medical forms, fragmented information spread across multiple healthcare systems, delays caused by prior authorization, and specialists who often appeared to begin their investigations with little visibility into years of prior evaluations, unsuccessful treatments, and clinical history.

Those experiences led me to ask a simple question: Could a national longitudinal patient history, combined with AI, reduce administrative costs while helping physicians make better-informed decisions?

The proposal outlined that vision, described the potential benefits, and acknowledged obvious challenges such as privacy, security, governance, and patient consent.

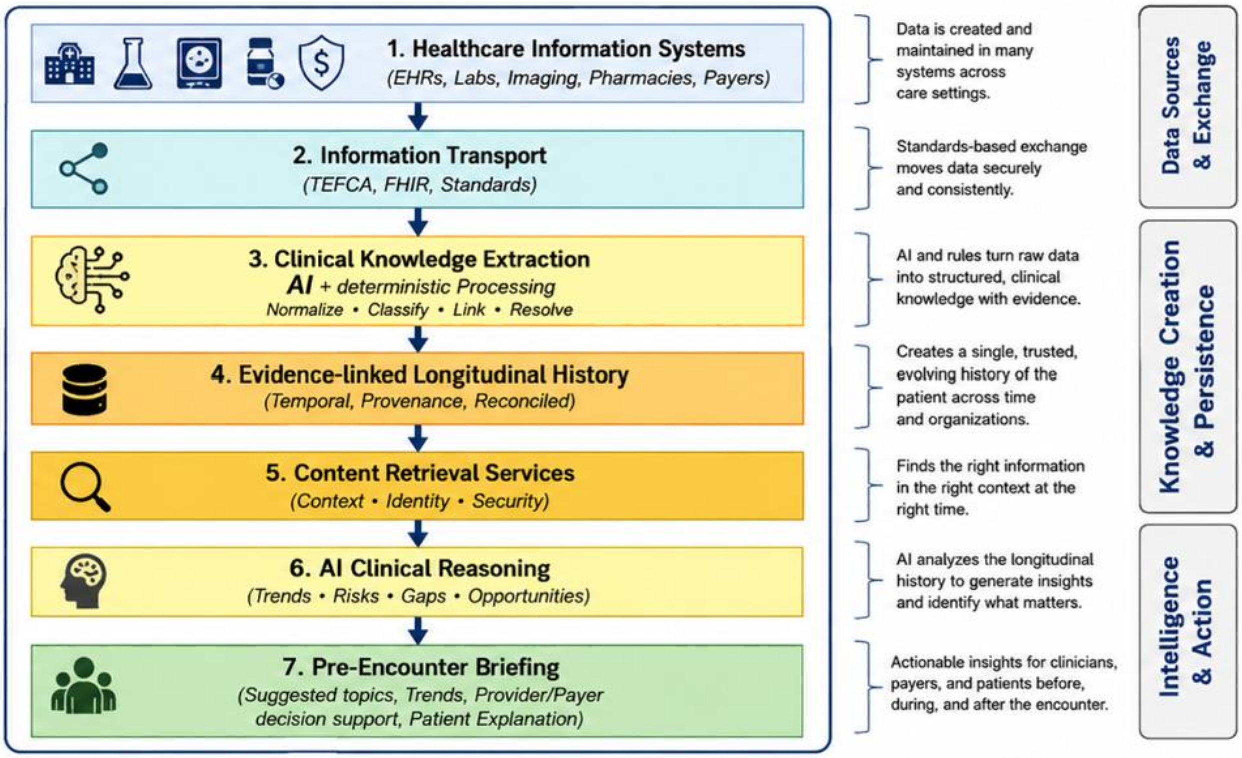


What I didn't fully appreciate was that AI is only one component of the solution. Building a trustworthy longitudinal patient history requires an information architecture that collects, transports, reconciles, preserves, retrieves, and reasons over healthcare information before AI can add meaningful value.

Rather than weakening my conviction, that exercise strengthened my appreciation for the scale of the problem and reinforced my belief that the idea deserves a structured feasibility study by experts from multiple disciplines.

Over the past several weeks, I've been thinking through each layer of that architecture. Every layer introduces its own technical and organizational challenges, and each deserves careful study before any implementation effort begins.

The Reality: An Information Architecture for AI-Enabled Healthcare



1. Healthcare Information Systems

Healthcare information doesn't arrive in one convenient format. It is distributed across EHRs, laboratories, imaging systems, pharmacies, and payers. Some information is highly structured (medications, laboratory values, procedures), while other information exists only in free-form clinical notes, imaging reports, scanned documents, or medical images. Any national longitudinal history must integrate all of these forms of information.

2. Information Transport

TEFCA and FHIR represent major advances in healthcare interoperability, but they are fundamentally designed for on-demand retrieval of information for an individual patient. A continuously maintained longitudinal history would likely require a robust change-data-capture architecture capable of publishing new, corrected, and deleted clinical information while detecting missed updates and recovering from communication failures.

3. Clinical Knowledge Extraction

Combining information from multiple organizations introduces ambiguity. One provider may record a diagnosis while another explicitly rules it out. Clinical notes may contradict structured diagnosis codes. The system must preserve provenance, supporting evidence, and temporal history while determining the current clinical interpretation rather than simply overwriting one answer with another.

4. Evidence-Linked Longitudinal History

A longitudinal history is more than a collection of records. It should preserve clinical assertions, supporting evidence, revisions, corrections, provenance, and the evolution of a patient's history over time. Technologies such as temporal databases may play an important role in preserving both clinical truth and historical understanding.

5. Context Retrieval

AI cannot reason effectively without context. The challenge is determining which portions of a patient's history are relevant to each new laboratory result, imaging study, physician note, or clinical question. Retrieving too little context risks missing important relationships, while retrieving everything would be inefficient and potentially overwhelming.

6. AI Clinical Reasoning

AI should not replace physician judgment. Instead, it should analyze the longitudinal history to identify significant trends, conflicting information, possible medication interactions, missing follow-up, and other issues that deserve clinical review. Every conclusion should remain linked to supporting evidence and explainable by both clinicians and patients.

7. Pre-Encounter Briefing

The objective is not another alert system. The goal is to prepare both physician and patient before the encounter by highlighting the few issues most deserving discussion, reducing the time spent searching for information and increasing the time available for clinical decision-making.

One conclusion has become increasingly clear to me: AI is not the architecture. AI is one component of a much larger information architecture. The real challenge is creating a trusted, evidence-linked, longitudinal patient history that provides AI with the context it needs to reason effectively.

That realization hasn't weakened my confidence in the vision. Quite the opposite. It has convinced me that the opportunity is even greater than I originally imagined—and that the engineering challenges deserve the attention of experts from healthcare, information architecture, interoperability, artificial intelligence, privacy, security, and public policy.

That's exactly why my proposal recommended a focused feasibility study rather than immediate implementation.